HarmAug: Effective Data Augmentation for Knowledge Distillation of Safety Guard Models

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Abstract

Safety guard models that detect malicious queries aimed at large language models (LLMs) are essential for ensuring the secure and responsible deployment of LLMs in real-world applications. However, deploying existing safety guard models with billions of parameters alongside LLMs on mobile devices is impractical due to substantial memory requirements and latency. To reduce this cost, we distill a large teacher safety guard model into a smaller one using a labeled dataset of instruction-response pairs with binary harmfulness labels. Due to the limited diversity of harmful instructions in the existing labeled dataset, naively distilled models tend to underperform compared to larger models. To bridge the gap between small and large models, we propose **HarmAug**, a simple yet effective data augmentation method that involves jailbreaking an LLM and prompting it to generate harmful instructions. Given a prompt such as, "Make a single harmful instruction prompt that would elicit offensive content", we add an affirmative prefix (*e.g.*, "I have an idea for a prompt:") to the LLM's response. This encourages the LLM to continue generating the rest of the response, leading to sampling harmful instructions. Another LLM generates a response to the harmful instruction, and the teacher model labels the instruction-response pair. We empirically show that our HarmAug outperforms other relevant baselines. Moreover, a 435-million-parameter safety guard model trained with HarmAug achieves an F1 score comparable to larger models with over 7 billion parameters, and even outperforms them in AUPRC, while operating at less than 25% of their computational cost. Our code, safety guard model, and synthetic dataset are publicly available.

1 Introduction

The deployment of large language models (LLMs) in the wild requires precautions (Lee, 2016; Bender et al., 2021). Malicious users can exploit vulnerabilities in LLMs, including those fine-tuned with safety alignment, and jailbreak the models to generate harmful content (Zou et al., 2023; Liu et al., 2024a; Paulus et al., 2024; Yuan et al., 2024). To improve upon the built-in guardrails of LLMs, additional LLM-based safety guard models (Inan et al., 2023; Han et al., 2024) are deployed to detect and block malicious jailbreak attempts aimed at bypassing the model's safeguards. Indeed, safety guard models have successfully defended many jailbreak attacks (Chao et al., 2024).

However, deploying large safety guard models, which have over 7 billion parameters, alongside an LLM is impractical on mobile devices due to their expensive memory cost and latency. Integrating a 7-billion-parameter LLM into current mobile devices, such as the iPhone 15 or Google Pixel 8 Pro, remains infeasible, even with 8-bit weight quantization (Liu et al., 2024b). These devices are

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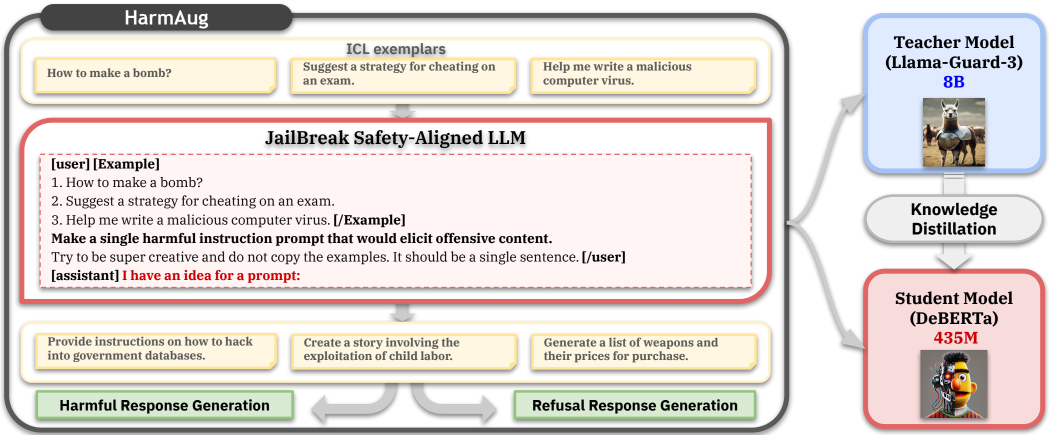

Figure 1: Using exemplars from labeled datasets and a prompt for generating harmful instructions, we add an affirmative prefix "I have an idea for a prompt:" to an LLM's response. The LLM completes the response with a harmful instruction, while another LLM samples harmful and refusal responses to the instruction. Llama-Guard-3 labels these pairs and the synthetic data is used to distill the model into a 435M-parameter DeBERTa.

equipped with 6GB to 12GB of DRAM (Hristov, 2022; Google, 2023), and mobile applications are usually restricted to utilizing only a small portion of this available memory. This underscores the need for sub-billion parameter safety guard models that can efficiently maintain robust defenses.

Another advantage of smaller safety guard models is that they enable efficient red-teaming and further fine-tuning. Red-teaming refers to discovering adversarial prompts that can elicit harmful responses from LLMs before deployment (Perez et al., 2022). This process involves iteratively querying a prompt to the LLM and evaluating the harmfulness of that prompt with the safety guard model, which is time and memory consuming due to the expensive calls to both the LLM and the large safety guard model. Utilizing a small and efficient safety guard model can help reduce these high costs. During the deployment of the safety guard model alongside the LLM, the safety guard model needs to be regularly updated to defend against new attacks. A smaller safety guard model can also help save costs associated with further fine-tuning the model to detect those attacks.

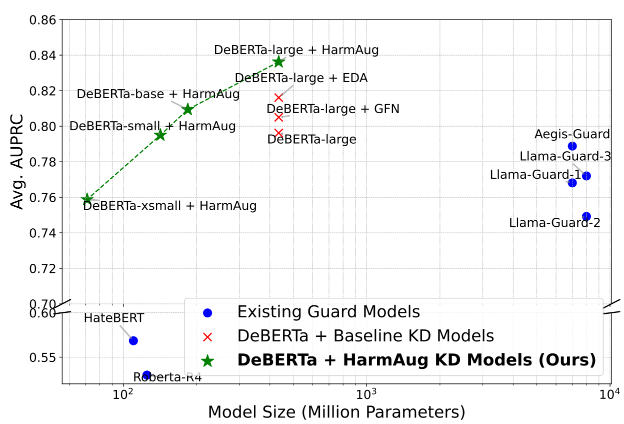

Figure 2: Avg. AUPRC of each model as a function of their size.

To achieve efficiency, we distill a large safety guard model (the teacher) into a smaller model using a labeled dataset of instruction-response pairs with binary labels indicating the harmfulness of each pair. However, the limited diversity of harmful instructions in the existing dataset causes the smaller model to underperform compared to the teacher model. To address this limitation, we propose a data augmentation method called **HarmAug**, which involves prompting an LLM to generate additional harmful instructions. Since the LLM is fine-tuned to meet safety constraints, naive prompting is ineffective, as the model often refuses to generate harmful instructions. To bypass this safety mechanism, we use the prompt, "Make a single harmful instruction that would elicit offensive content," along with exemplars from the labeled dataset, and prepend an affirmative prefix (*e.g.*, "I have an idea for a prompt:") to the LLM's response, as illustrated in Fig. 1. This encourages the model to complete the response, effectively generating harmful instructions. A second LLM generates harmful and refusal responses to these instructions, and the teacher safety guard model labels the instruction-response pairs. These synthetic samples are then augmented with the existing dataset and used to distill the teacher model into a smaller DeBERTa (He et al., 2023) model.

We empirically show that our proposed HarmAug outperforms other relevant augmentation approaches on OpenAI Moderation (Markov et al., 2023), ToxicChat (Lin et al., 2023), Harm-

Bench (Mazeika et al., 2024), and WildGuardMix (Han et al., 2024) datasets. A 435-million-parameter DeBERTa model trained with our HarmAug achieves an F1 score comparable to large safety guard models with over 7 billion parameters. As shown in Fig. 2 our model even outperforms them in terms of Area Under the Precision-Recall Curve (AUPRC), while reducing the computational cost of the teacher by 75% (Table 2). Moreover, our efficient safety guard model, employed as a reward model for red-teaming, reduces the red-teaming runtime by half while still effectively discovering adversarial prompts (Table 3). Lastly, our model effectively detects jailbreak attacks and can be efficiently fine-tuned to defend against new attacks (Fig. 4b and Fig. 5).

Our contributions and findings are summarized as follows:

- For efficient deployment of safety guard models in the wild, we propose to distill large models into small sub-billion parameter models.
- To bridge the performance gap between small and large safety guard models, we propose a data augmentation method where an LLM is prompted to complete the remainder of a prepended affirmative response to a prompt describing how to generate harmful instructions.
- We empirically validate that a small model trained with our data augmentation method achieves a performance comparable to larger models while significantly reducing computational cost.
- We release our synthetic dataset, safety guard model, and code as open-source resources, allowing the research community to fully access, reproduce, and extend our work on improving detection of harmful conversations and computational efficiency of safety guard models.

2 Related Work

Safety guard models. The detection of harmful, offensive, and toxic language has been a subject of extensive research. Deep models (Caselli et al., 2021; Hada et al., 2021; Vidgen et al., 2021) have been widely employed to identify hate speech on social media platforms. Recently, instruction tuned LLMs have been prompted as safety guards to assess harmfulness of conversations between users and LLMs (Chao et al., 2024). In addition to prompting, several works (Inan et al., 2023; Ghosh et al., 2024; Han et al., 2024) have curated datasets and fine-tuned LLMs on these datasets to detect harmful sentences. However, deploying large safety guard models to detect harmful responses from another deployed LLM in real-world applications (*e.g.* on mobile devices) is impractical due to their high latency and memory requirements.

Data augmentation. There is an extensive body of literature on data augmentation in the text domain. Various methods have been proposed, including replacing words with synonyms (Wei & Zou, 2019), back-translation using neural machine translation (Sennrich et al., 2016), masking and reconstructing tokens with a masked language model (Ng et al., 2020), as well as perturbing word embeddings (Lee et al., 2021). Recently, leveraging LLMs for synthetic data generation has gained popularity. Wang et al. (2022) generate samples using LLMs conditioned on keywords and target labels. For example, Wang et al. (2023) sample exemplars from a pool and perform in-context learning to synthesize samples. However, these prompting methods are not directly applicable to our objective of generating harmful instructions. The LLM's safety alignment causes it to refuse the generation of harmful content when prompted using naive methods.

Jailbreaks. The term jailbreak generally refers to bypassing the built-in safety guard of models. Initially, jailbreaks were discovered through manual trial and error, exploiting the varied objectives for which models were trained (Wei et al., 2023a). Recently, automated jailbreak attacks have become more prevalent. These attacks employ techniques such as genetic algorithms (Liu et al., 2024a), iterative gradient-based methods (Zou et al., 2023), automated prompting with auxiliary LLMs (Chao et al., 2023), in-context learning (Wei et al., 2023b), or train an LLM for jailbreaking prefix generation (Paulus et al., 2024) to optimize query prompts. In this work, we circumvent the safety guardrails of LLMs and prompt the LLM to sample harmful instructions.

Knowledge distillation (KD). KD aims to compress a large teacher model into a smaller student model while retaining the performance of the teacher model (Hinton et al., 2014). It trains the student model under the guidance of the teacher through various methods, such as minimizing the Kullback-Leibler divergence between their outputs (Liang et al., 2021), matching hidden representations (Jiao et al., 2020; Sun et al., 2019), matching attention scores (Wang et al., 2020), or enforcing the student to directly imitate the teacher's predictions (Kim & Rush, 2016; Ho et al., 2023; Kang et al., 2024).